

1 Question (14 points)

Given n numbers (positive or negative) representing the genes in the chromosomes (Example: +1 -2 -4 +3)

- a) How many permutations are there for those n elements?
- b) How many permutations of these have two breakpoints?

2 Question (15 points)

Suppose that you are given an array A of n words sorted in lexicographic order and want to search this list for some arbitrary word, perhaps w (we write the number of characters in w as $|w|$). Design three algorithms to determine if w is in the list: one should have $O(n|w|)$ running time; another should have $O(|w|\log n)$ running time but use no space (except for A and w); and the third should have $O(|w|)$ running time but can use as much additional space as needed.

3 Question (16 points)

Design two algorithms for finding the longest string shared by two given texts of length n .

- a) The first algorithm will have $O(n^3)$ time complexity and compares substrings of two texts one by one.
- b) The second algorithm will have $O(n^2)$ time complexity and uses Suffix trees.

4 Question (15 points)

Suppose that the task is to cluster the following eight points (with (x, y) representing location) into three clusters. $A_1(2, 10)$, $A_2(2, 5)$, $A_3(8, 4)$, $A_4(5, 8)$, $A_5(7, 5)$, $A_6(6, 4)$, $A_7(1, 2)$, $A_8(4, 9)$. The distance function is Euclidian distance. Suppose initially we assign A_2 , A_5 , and A_8 as the center of each cluster, respectively. Run the k-means algorithm one epoch on this data.

5 Question (20 points)

Let us say we have 2 states:

S_1 : α , S_2 : β with equal initial probabilities.

and the transition matrix is:

$$\mathbf{A} = \begin{vmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{vmatrix}$$

The emission probabilities for S_1 is:

$$\mathbf{e}(S_1) = \begin{vmatrix} \text{A: } \frac{2}{5} \\ \text{G: } \frac{2}{5} \\ \text{T: } \frac{1}{10} \\ \text{C: } \frac{1}{10} \end{vmatrix}$$

The emission probabilities for S_2 is:

$$\mathbf{e}(S_2) = \begin{vmatrix} \text{A: } \frac{1}{5} \\ \text{G: } \frac{1}{5} \\ \text{T: } \frac{3}{10} \\ \text{C: } \frac{3}{10} \end{vmatrix}$$

Decode the most likely sequence of states for sequence GGCT.

6 Question (20 points)

Construct unrooted phylogenetic tree from the following distance matrix:

	A	B	C	D	E	F
A	0	37	55	69	68	25
B	37	0	42	56	55	38
C	55	42	0	46	45	56
D	69	56	46	0	25	70
E	68	55	45	25	0	69
F	25	38	56	70	69	0

A, B, C, D, E, and F are the names of the species.